SOFTOMATIC
TECHNOLOGY SERVICES

EXAMATE · SYSTEM DESIGN · ELITE + BATCH 1

Lesson 4 · Back-of-the-Envelope Estimation

Turn "a lot of users" into numbers that decide the design. Told through one story: an interviewer asks Asha a single question, and every box on her whiteboard depends on the answer.

MODULE
System Design · Part A
Foundations

TIME
About 80 minutes
plus 50 minutes of practice problems

YOU NEED
Multiply and divide
by powers of ten. That's it.

YOUR ROUTE TODAY

01 The 9,000 question
02 Units, QPS and peak
03 Storage, bandwidth, cache
04 Latency and servers
05 **Numbers into decisions**

LESSON 4 AT A GLANCE

What you will be able to do

Estimation is **reasoned arithmetic from stated assumptions**. Its only job is to size the problem so you choose the right design.

Users into QPS
Average and peak QPS from users and actions, in under a minute.

Storage, bandwidth, cache
With retention and replication, each number said with its unit.

Compare designs with latency
Use latency numbers to compare two designs without building either.

QPS into servers
A rough server count, sanity-checked against something real.

Estimates that decide
Know when a number changes the design and when it is decoration.

Awkward cases
No scale given, a disagreeing interviewer, or no estimation wanted.

Subject / module	System Design / Part A: Foundations
Difficulty	Beginner. If you can multiply and divide by powers of ten, you can do this.
Prerequisites	Lessons 1 to 3. Lesson 1 introduced QPS and an early estimate; this is the full toolkit.
Next lesson	Lesson 5: Scaling, load balancers, DNS and proxies
Companies covered	Amazon, Microsoft, Google, Cisco, Adobe, Atlassian, Uber, Flipkart

SOURCE-CONFIRMED
A dated first-person report naming the company and the question, Oct 2024 to Oct 2026, opened in full.

CANDIDATE-REPORTED
Compilations, aggregators, forum threads, older than 24 months, undated, or seen only in search extracts.

PATTERN-ONLY
Commonly asked, but no company source, so no company name is attached.

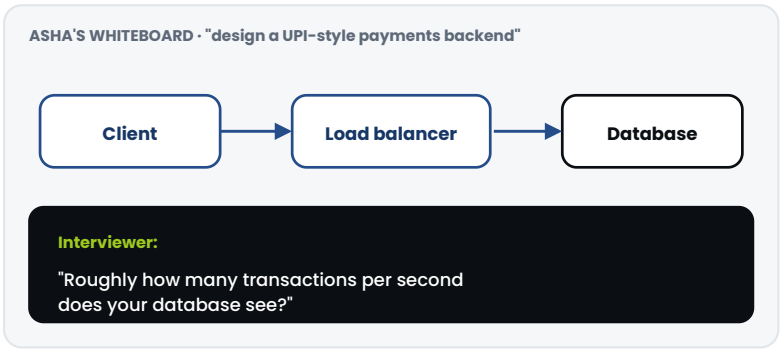
AN HONEST NOTE ON HOW MUCH THIS MATTERS

Estimation is the **least consistently graded** part of a design round. Cisco interviewers asked about QPS in a July 2026 whiteboard round [S6]. Anonymous Blind posters disagree: some say Google and Facebook "definitely" want calculations, others aced design rounds with "not a single calculation" [S5]. A guide says juniors may be forgiven weak estimation if the architecture is solid [X4]. So learn to do it fast, and **ask whether it is wanted**. It is worth about an hour of practice, not a week.

ONE QUESTION, ONE WHITEBOARD

Asha and the 9,000 question

Asha is asked to design the backend for a UPI-style payments app. She draws a client, a load balancer and a database. Then the interviewer asks one question.



THREE ANSWERS

- "A lot" — tells them nothing ✗
- "About a million per second" — about 100x too high ✗
- "About 9,000 per second" — 791 million a day / 86,400 ✓

Source: NPCI data as reported 1 Sep 2026 [S7]

Every box is sized by the answer. Off by 100x, and the load balancer, the database and the server count are all wrong.

Here is a real anchor. NPCI's data, as reported on 1 September 2026, shows UPI processed **24.51 billion transactions in August 2026**, an average of **791 million per day** [S7]. Can you turn "791 million per day" into "per second" in your head?

PAUSE AND PREDICT

Roughly how many transactions per second is that on average? Closer to 900, 9,000 or 90,000?

Answer: About 9,000 (791,000,000 / 86,400 = 9,155). If you said 900, the system would melt at 10:01 on the first of the month, when everyone remembers their rent.

INTERVIEW MODE: SAY IT LIKE THIS

- An estimate turns vague scale into numbers that decide the design.
- State the assumption, show the arithmetic, round to the right order of magnitude.
- Off by 2x is fine. Off by 100x changes the architecture.

MEMORY HOOK: THE ONE RULE

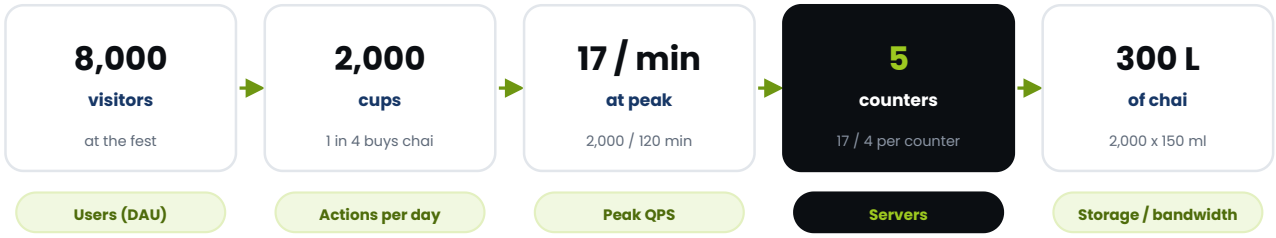
A number without its **assumption** and its **unit** is not an estimate. "9,000" is noise; "about 9,000 transactions per second, assuming an even day" is an estimate.

BEFORE THE JARGON

Planning the food for a college fest

Your fest expects 8,000 visitors. The food committee must decide how many chai counters to set up. Nobody builds a counter first. They do napkin maths.

NAPKIN MATHS FOR THE COLLEGE FEST



Bottom row: the same idea in system design. Nobody predicted the exact crowd; they picked assumptions and ended with a decision.

Five steps, one decision. They did not predict the exact crowd. If the real answer is 4 or 6 counters, nobody suffers; 50 wastes money, 1 starts a queue riot.

AT THE FEST	IN SYSTEM DESIGN
Visitors	Users (DAU)
Chai buyers per visitor	Actions per user per day
Rush window	Peak factor
Cups per minute	QPS
Counters needed	Servers needed
Litres of chai	Storage and bandwidth

WHAT GOES WRONG

Where the analogy breaks. Chai does not copy itself, but data does: you store three copies for safety, so storage needs a multiplier the chai stall never had. And a counter is open or closed, while a server slows down smoothly as it gets busier, so a server count always includes headroom.

TINY CHALLENGE

12,000 visitors, 1 in 3 buys chai, rush of 3 hours. Cups per minute at peak? Answer: 4,000 cups / 180 min = about 22 per minute.

INTERVIEW MODE: SAY IT LIKE THIS

- I convert users into requests per day, divide by 86,400 for average QPS, multiply by a peak factor, and compare with what one server handles.

EVERY WORD, ONE LINE

Vocabulary primer

CDN, load balancer and sharding are only named here; Lessons 5 to 7 cover them.

Estimate
A reasoned approximate number built from stated assumptions.

Back-of-the-envelope
Quick, rough arithmetic you could do on an envelope.

Order of magnitude
The power of ten of a number: 900 and 9,000 differ by one.

Assumption
A number you choose and say aloud because you were not given it.

DAU
Daily active users: distinct people who use the system in one day.

QPS (RPS)
Queries (requests) per second.

Average vs peak QPS
Average = per day / 86,400; peak = busiest moment, often a multiple.

Peak factor
The multiplier from average to peak (an assumption: 2x, 3x, 10x).

Read:write ratio
Reads per write. 100:1 means 100 reads for every write.

Payload
The size of the data in one request or reply, in bytes.

Throughput / bandwidth
Data moved per second: MB/s (bytes) or Mbps (bits).

Byte / bit
A byte is 8 bits; networks are quoted in bits, storage in bytes.

KB, MB, GB, TB, PB
About 10^3, 10^6, 10^9, 10^12, 10^15 bytes.

Powers of two
Computers count in 2^n; 2^10 = 1,024, about one thousand [S3].

Retention
How long data is kept (30 days, 5 years).

Replication factor
How many copies of each piece of data are stored (Lesson 7).

Latency
Time for one operation, in ns, µs or ms.

ns, µs, ms
Nanosecond 10^-9 s, microsecond 10^-6 s, millisecond 10^-3 s.

RTT
Round-trip time for a message to go and come back (Lesson 2).

Working set / hot data
The part of the data read often; what a cache should hold.

Cache
Fast memory holding copies of hot data (Lesson 6).

Headroom
Spare capacity for spikes and failures, so you never run at 100%.

Concurrency
How many requests are in flight at the same moment.

Egress
Data leaving your system toward users; often billed by cloud providers.

Sanity check
Comparing your estimate with something known to see if it is plausible.

INTERVIEW MODE: SAY IT LIKE THIS

- Attach a unit to every number: "9,000 requests per second", "3 TB over five years", "15 megabits per second". A bare number is not an estimate.

CONCEPT 1 · WHY ESTIMATE AT ALL?

Compare designs without building them

A design that is perfect for 10 requests per second is wrong at 10,000. You cannot choose until you know how big the problem is.

Jeff Dean advocates evaluating designs with back-of-the-envelope calculations: estimates built from thought experiments and common performance numbers, so you can judge which design meets your requirements without building it [S1]. His example: an image-results page of **30 thumbnails**.



Serial vs parallel reads, numbers from the source [S1].

DESIGN	CALCULATION	TIME
Serial: seek, read a 256 KB image, repeat 30 times	30 seeks x 10 ms + 30 x 256 KB / 30 MB/s	about 560 ms
Parallel: issue all 30 reads at once	10 ms seek + 256 KB / 30 MB/s	about 18 ms (the source says 30 to 60 ms is more likely, because of variance)

Check the serial figure: 30 x 10 ms = 300 ms, and 30 x 256 KB = 7.5 MB, which at 30 MB/s is about 250 ms, so about 550 to 560 ms in total.

WHAT GOES WRONG

Spending ten minutes on arithmetic nobody asked for, or skipping it when the interviewer expected numbers. Like cooking for the fest without asking how many are coming: you feed forty or four thousand.

TINY CHALLENGE

In Dean's example, which single number makes the serial design slow? Answer: The 30 separate 10 ms seeks, 300 ms in total.

INTERVIEW MODE: SAY IT LIKE THIS

- "Before I pick components, I would like to size the problem. Would you like me to estimate traffic and storage, or should I assume a scale?"
- "I will round aggressively and state my assumptions; tell me if a number looks off."

CONCEPT 2 · UNITS, POWERS OF TWO, TIME

Most errors are unit slips

KB vs MB, per day vs per second, bits vs bytes. One small table prevents almost all of them [S3].

BYTES: EACH STEP UP IS x 1,000

1 TB	`2^40 = 1,099,511,627,776`	about one trillion (10^12)
1 GB	`2^30 = 1,073,741,824`	about one billion (10^9)
1 MB	`2^20 = 1,048,576`	about one million (10^6)
1 KB	`2^10 = 1,024`	about one thousand (10^3)

1 MB = 8 Mb: storage is quoted in bytes (B), network links in bits (b).

TIME IN SECONDS

1 minute	60
1 hour	3,600
1 day	86,400 (about 10^5)
1 month (30 d)	2,592,000
1 year (365 d)	31,536,000

Use the decimal words (thousand, million, billion, trillion) and move up x 1,000 per step: KB, MB, GB, TB, PB.

Shortcut 1. 1 million requests per day is about **12 per second** (1,000,000 / 86,400 = 11.6).

Shortcut 2. A prep guide suggests dividing by 100,000 instead of 86,400 [X1]. That underestimates by 1 - 86,400/100,000 = **13.6%**, fine at the order-of-magnitude level.

100 MILLION REQUESTS A DAY	RESULT
Exact: 100,000,000 / 86,400	1,157 per second
Shortcut: 100,000,000 / 100,000	1,000 per second (13.6% low)

WHAT GOES WRONG

Treating 1 GB as 10^6 bytes; mixing MB and Mb (a megabyte is 8 megabits); dividing a monthly number by 86,400 instead of 2,592,000.

MEMORY HOOK: B FOR BIG

Big **B** = **Byte** = 8 small b's (bits). Disks speak Bytes, wires speak bits.

TINY CHALLENGE

50 million requests a month: how many per second? Answer: 50,000,000 / 2,592,000 = about 19.3.

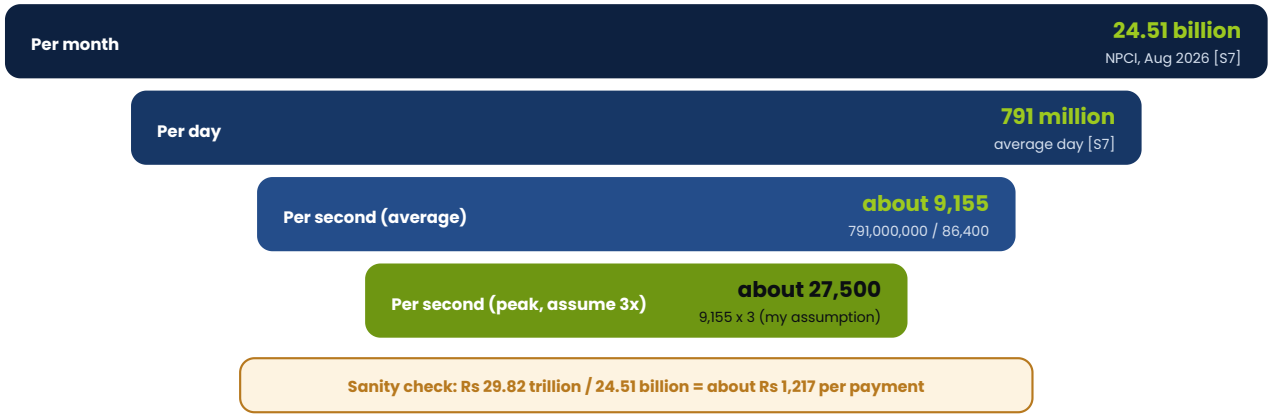
INTERVIEW MODE: SAY IT LIKE THIS

- "A day is 86,400 seconds, so a million requests a day is about twelve per second."
- "I use thousand, million, billion, trillion for KB, MB, GB, TB, and keep bits and bytes separate."

CONCEPT 3 · FROM USERS TO QPS

The headline is users; the load is QPS

The chain: DAU x actions per user per day = requests per day; / 86,400 = average QPS; x peak factor = peak QPS.



Back to Asha. Real data, one division, one stated assumption. The 3x peak is mine; the source gives no peak figure.

A social feed [S4]. 200 million DAU, each writing 2 posts and reading about 200 a day: 400 million writes a day is about **5,000/s**, 40 billion reads a day about **500,000/s**, a read:write ratio of **100:1**.

Check: 400,000,000 / 86,400 = 4,630 and 40,000,000,000 / 86,400 = 462,963. The guide says interviewers want a defensible order of magnitude, not 4,629 versus 5,000 [S4].

LINE (UPI, AUG 2026)	CALCULATION	RESULT
Average rate	791,000,000 / 86,400	about 9,155/s
Peak (assume 3x)	9,155 x 3	about 27,500/s
Average value (sanity)	Rs 29.82 trillion / 24.51 billion	about Rs 1,217

About Rs 1,200 per payment is plausible for chai, groceries and rides [S7 gives the totals].

PAUSE AND PREDICT

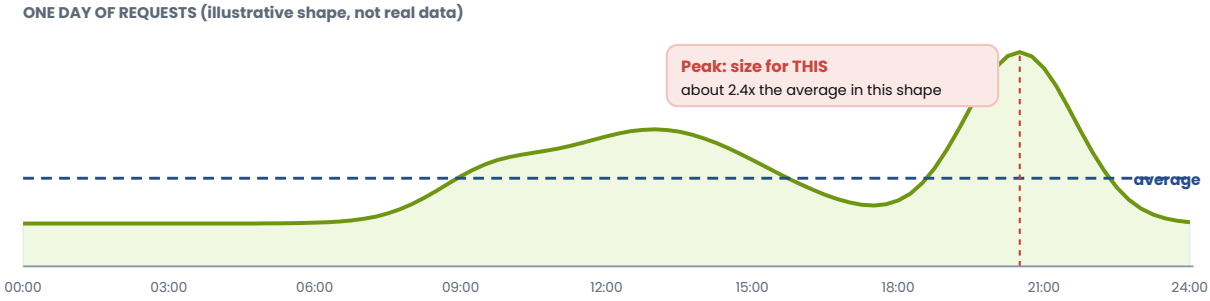
Why not size servers for the average?

Answer: Because systems fail at the peak. The average hides the 12:00 sale and the evening rush.

AVERAGE HIDES SPIKES

The peak factor is an assumption you say aloud

Say "I will assume peak is 3 times average" and give a reason: evening usage, a sale at 12:00. The right value comes from data or from asking.



Illustrative shape. The dashed line is what averages tell you; the red line is what crashes you.

Worked example 2: our short-link service (the running system from Lessons 1 to 3). 100 million new links a month and 100 reads per write.

LINE	CALCULATION	RESULT
Creates per second (average)	100,000,000 / 2,592,000	about 38.6
Redirects per second (average)	38.6 x 100	about 3,860
With a 2x peak	38.6 x 2 and 3,860 x 2	about 77 and 7,700 per second

WHAT GOES WRONG

Sizing capacity with the average, then falling over at the daily spike. IRCTC Tatkal at 10:00 sharp is the national lesson in why "average" is a lie. Also dropping the read:write ratio: it decides cache and replica choices later.

TINY CHALLENGE

30 million DAU, 10 actions each: average QPS?
Answer: 300,000,000 / 86,400 = about 3,470.

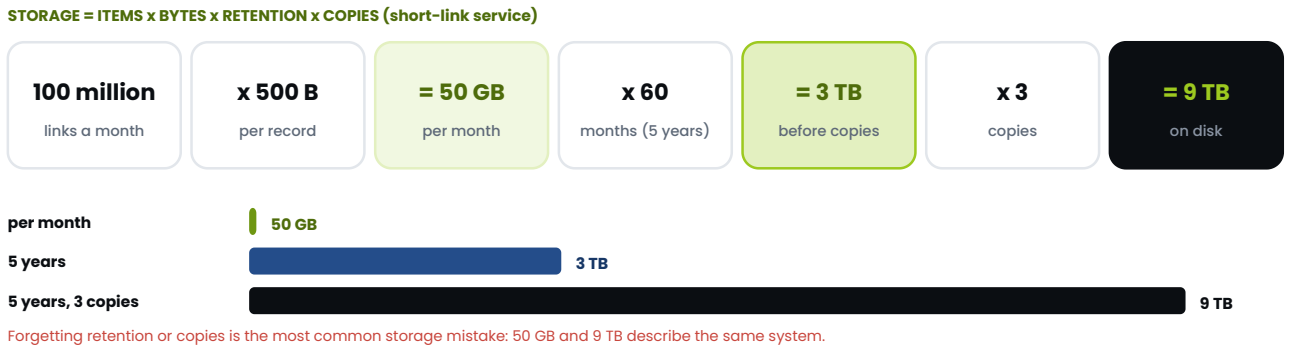
INTERVIEW MODE: SAY IT LIKE THIS

- "DAU times actions per day gives requests per day; divide by 86,400 for average QPS; multiply by my peak assumption."
- "Reads outnumber writes about 100 to 1, so I will optimise the read path."

CONCEPT 4 · STORAGE

Items x bytes x retention x copies

"How big will the database be?" decides whether one machine is enough.



Four multipliers. Leave out any one and the answer is wrong by that factor.

Size a record by adding fields. A published worksheet sizes a URL-shortener record as short URL (7 B), original URL (about 100 B), timestamp (8), user ID (8) and metadata (about 50), roughly 200 B [X2]. The fields add to 173 B, so "about 200" is a fair round-up. Real systems add index and row overhead, which is why Lessons 1 to 3 used 500 B: **state whichever you choose.**

5 YEARS	200 B/RECORD	500 B/RECORD
Per month	20 GB	50 GB
Records (60 months)	6 billion	6 billion
Storage	1.2 TB	3 TB
With 3 copies	3.6 TB	9 TB

BONUS CHECK

6 billion links need 6 billion distinct codes. A 7-character code from 62 symbols has 62^7 = **3,521,614,606,208** (about 3.5 trillion) values, so 6 billion uses only **about 0.17%** of the space. Lesson 24 builds on this.

UPI scale (assumption flagged): if each transaction record is 1 KB, 791 million a day x 1 KB = 791 GB a day; x 365 = about 289 TB a year; x 3 copies = about 866 TB, close to 0.9 PB.

MEMORY HOOK: I BOUGHT RICE CHEAP

Items x **B**ytes x **R**etention x **C**opies.

TINY CHALLENGE

10 million messages a day at 200 B, kept 1 year, 3 copies? Answer: 2 GB/day; x 365 = 730 GB; x 3 = 2.19 TB.

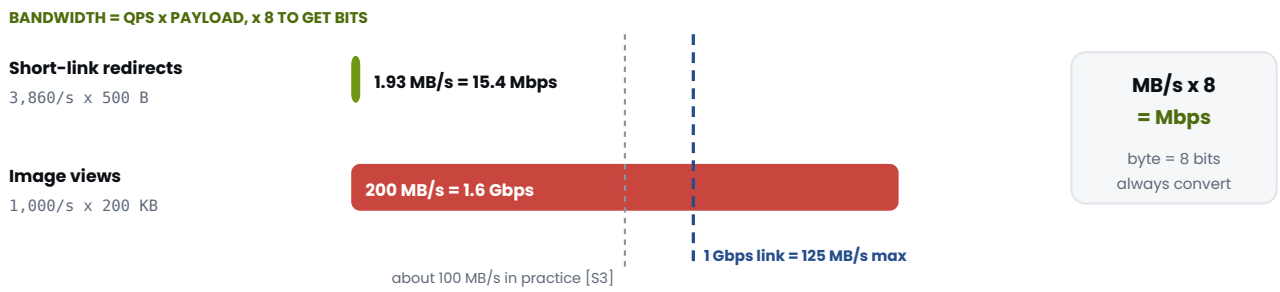
INTERVIEW MODE: SAY IT LIKE THIS

- "Storage is items per day, times bytes per item, times retention, times replication."

CONCEPT 5 · BANDWIDTH

Storage can fit while the network cannot

Bandwidth = QPS x payload bytes. Multiply by 8 to compare with network links, which are quoted in bits per second.



Handy figures [S3]: about 100 MB/s over 1 Gbps Ethernet, about 1 GB/s from SSD, about 4 GB/s from main memory. 1 Gbps is 125 MB/s in theory; 100 MB/s is the practical figure.

EXAMPLE	CALCULATION	RESULT
Short-link redirects, average	3,860 x 500 B = 1.93 MB/s; x 8	15.4 Mbps
Short-link redirects, peak	7,720 x 500 B = 3.86 MB/s; x 8	about 31 Mbps: tiny
Image views	1,000 x 200 KB = 200 MB/s; x 8	1.6 Gbps: more than one 1 Gbps link, before peak

The image number is the kind that pushes you toward a **CDN** (Lesson 5) and toward watching egress bills.

WHAT GOES WRONG

Reporting MB/s as Mbps (an 8x error), or forgetting outbound data to users is usually larger than inbound. A comment on the latency gist adds: "1 Gbps" is a bandwidth, not a latency, so a row mixing them misleads [S2].

TINY CHALLENGE

500 requests/s at 2 KB each, in Mbps? Answer: 500 x 2 KB = 1 MB/s = 8 Mbps.

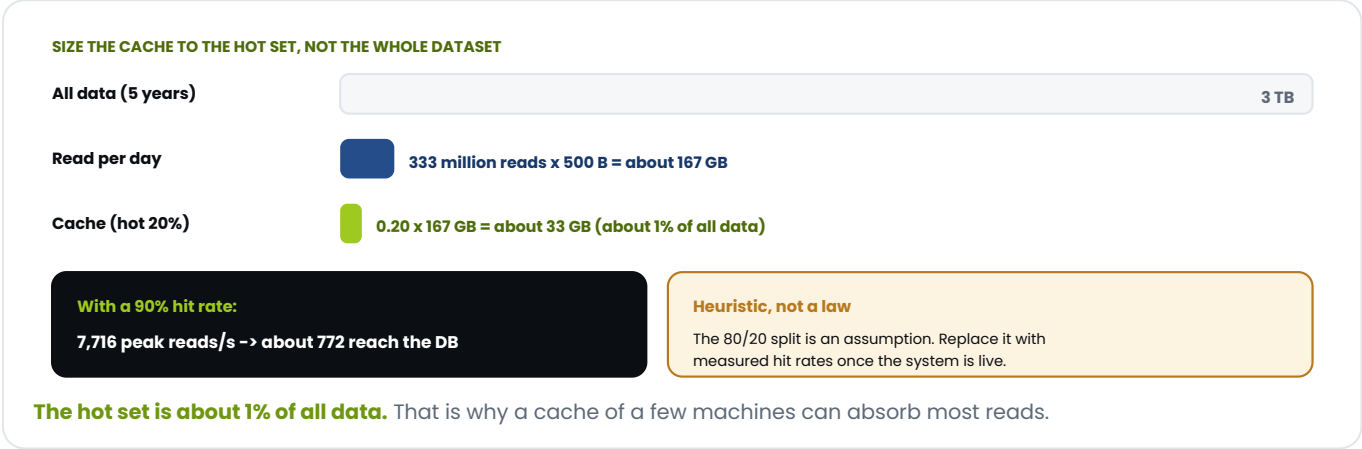
INTERVIEW MODE: SAY IT LIKE THIS

- "Bandwidth is QPS times payload size, multiplied by eight to get bits."
- "Redirects are about 15 megabits per second; serving images is what would need a CDN."

CONCEPT 6 · MEMORY AND CACHE SIZING

Cache the hot set, not everything

Estimate the daily read volume in bytes and cache a fraction of it. A common heuristic caches the hottest 20% of daily read traffic.



The hot set is about 1% of all data. That is why a cache of a few machines can absorb most reads.

LINE	CALCULATION	RESULT
Reads per day	3,858 per second x 86,400	about 333 million
Bytes read per day (500 B each)	333,000,000 x 500 B	about 167 GB
Cache the hottest 20%	0.20 x 167 GB	about 33 GB
DB reads at peak, 90% hit rate	7,716 x 0.10	about 772 per second

33 GB fits in the memory of one large machine (a 64 GB machine is an assumption) or a few smaller ones. The 80/20 rule is a heuristic not opened in a source here; replace it with measured hit rates in real systems.

WHAT GOES WRONG

Sizing the cache to the whole dataset, or ignoring that the hot set changes over time. Cached data has overhead, and Lesson 6's eviction policy decides what stays.

TINY CHALLENGE

100 GB of reads a day, cache the hottest 20%?
Answer: 20 GB.

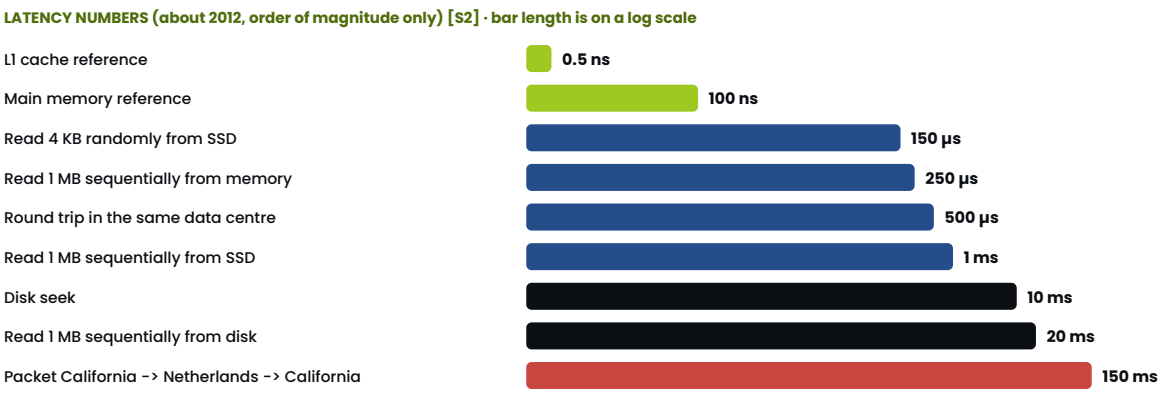
INTERVIEW MODE: SAY IT LIKE THIS

- "I size the cache to the hot subset, for example the most-read 20% of daily traffic, about 33 gigabytes here."
- "With a 90% hit rate the database sees about a tenth of the reads."

CONCEPT 7 · LATENCY NUMBERS

Every boundary costs orders of magnitude

The widely shared table credited to Peter Norvig and Jeff Dean (about 2012), as kept in a popular gist [S2].



Memory -> data-centre round trip: 5,000x · round trip -> disk seek: 20x · round trip -> across continents: 300x

Notice: every time you cross a boundary (memory, network, disk, continent) the cost jumps one to three orders of magnitude.

SHORT-LINK PATH	STEPS	TIME
Cache hit	app to cache RTT (0.5 ms) + memory read (0.0001 ms)	about 0.5 ms
Cache miss	above + app to DB RTT (0.5 ms) + SSD random read (0.15 ms)	about 1.15 ms

Both are about a millisecond. A user's own network round trip is often 20 to 100 ms (assumed), so backend time is small next to it. That tells you where speed-ups matter.

CAVEATS TO STATE IF CHALLENGED

The table is about 2012. Versions differ: the Google SRE-book variant reads 1 MB from disk in 5 ms, Dean's original lists 30 ms [S1, S3, X7]. Commenters note it ignores queueing and contention, and SSDs vary [S2]. A community 2026 edition adds LLM rows (about 1,000 ms to first token, about 3,000 ms for a short response) [S2]: rough and unvetted. Use it for **order of magnitude** only.

TINY CHALLENGE

How many times slower is a disk seek than a data-centre round trip? Answer: 10 ms / 0.5 ms = 20 times.

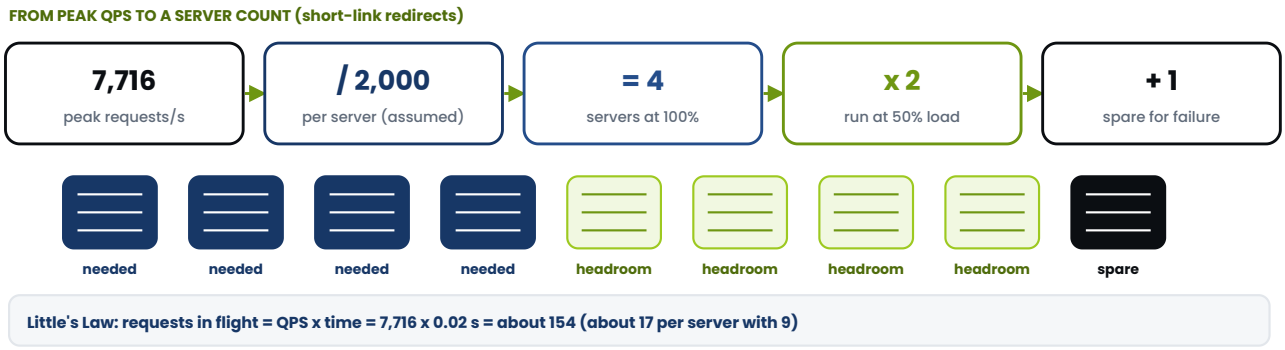
INTERVIEW MODE: SAY IT LIKE THIS

- "Memory is nanoseconds, a data-centre round trip half a millisecond, a disk seek ten milliseconds, a cross-continent round trip about 150 milliseconds."

CONCEPT 8 · FROM QPS TO A SERVER COUNT

Peak / per-server, plus headroom and a spare

Per-server QPS is an assumption you state (or measure with a load test such as wrk, Lesson 2).



Little's Law (concurrency = arrival rate x time in system) is a standard queueing result; no source was opened for it in this lesson.

LINE	CALCULATION	RESULT
One app server handles (assumption)	simple redirect	2,000 requests/s
Servers at peak	7,716 / 2,000	3.9, so 4
Run at 50% load for headroom	4 x 2	8
One spare for failure	8 + 1	9
Requests in flight, 20 ms each	7,716 x 0.02 s	about 154
Per server (9 servers)	154 / 9	about 17

The database sees about 772 reads per second after a 90% cache hit and about 77 writes per second at peak, which one node can handle (assumption: a node handles thousands of simple queries per second).

WHAT GOES WRONG

Quoting a server count without the per-server assumption, running at 100% (no room for spikes), or forgetting that replicas and spares cost money too.

TINY CHALLENGE

Peak 12,000 QPS, 3,000 per server: servers at 50% load? Answer: 12,000 / 3,000 = 4; at 50% load, 8.

INTERVIEW MODE: SAY IT LIKE THIS

- "Servers equal peak QPS divided by per-server capacity, with headroom and a spare. I assume two thousand simple requests per second per server and would confirm with a load test."

CONCEPT 9 · SANITY CHECKS

A confident wrong number is worse than a rough right one

Compare with something real

If your system claims 1 billion transactions a day, remember UPI's whole national volume is about 791 million a day [S7]. Either you are bigger than UPI or an assumption is off.

Check units

MB vs Mb, per day vs per second, KB vs MB.

Check completeness

Peak, retention, replication and the read:write ratio: all four in?

Say assumptions, invite correction

"Assuming 30 million daily users, I get about 3,500 requests per second. Does that match what you have in mind?"

TIME-BOX IT

DesignGurus puts clarify-and-scope, including scale estimates, in roughly the **first 5 to 10 minutes** [S4]. Estimation is a few minutes inside that. A five-step worksheet many guides use: clarify scale, traffic (average and peak), storage, bandwidth, then memory and servers [X2].

A slip caught by a sanity check. You estimate image bandwidth as 200 MB/s and write "200 Mbps". That is an 8x error: 200 MB/s x 8 = 1,600 Mbps = 1.6 Gbps.

MEMORY HOOK: THE WORKSHEET ORDER

Traffic, **S**torage, **B**andwidth, **C**ache, **S**ervers: "**The Smart Batch Cracks Systems**".

WHAT GOES WRONG

Going silent while calculating, or defending a number the interviewer has corrected. Arguing 9,155 versus 9,000 is the fastest way to spend five minutes on nothing.

TINY CHALLENGE

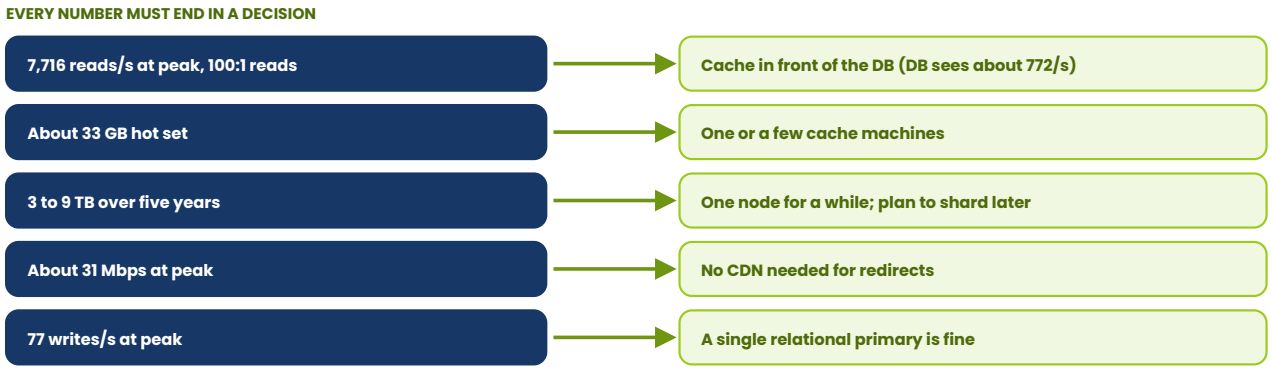
Spot the error: "5 million requests a day is 5,000,000 / 3,600 = 1,389 per second." **Answer: Wrong divisor: a day has 86,400 seconds, so about 58 per second.**

CONCEPT 10 · WHEN AN ESTIMATE CHANGES THE DESIGN

If no box changes, it was decoration

Typical translations (rules of thumb that depend on hardware; confirm with measurement).

IF THE ESTIMATE SHOWS...	THE DESIGN RESPONSE	LESSON
Reads far exceed writes (100:1)	Add a cache and read replicas	6, 7
Hot data is a small fraction of total	Cache only the hot set	6
Peak write rate far above one node	Queue the writes, batch, or shard	7, 10
Total data far above one node's disk	Shard the data	7
Outbound bandwidth above a link or budget	CDN, compression, more links	5
Latency budget smaller than the sum of hops	Fewer round trips, parallel calls	2, 5
Retention makes storage huge	Tier or archive old data	7



The short-link service, from numbers to decisions.

WHAT GOES WRONG

Computing everything, then drawing the diagram you would have drawn anyway. Or over-reacting: sharding a 100 GB database "because scale".

TINY CHALLENGE

1,000 writes/s and the primary handles 500: two responses? Answer: Queue and batch the writes; shard the table.

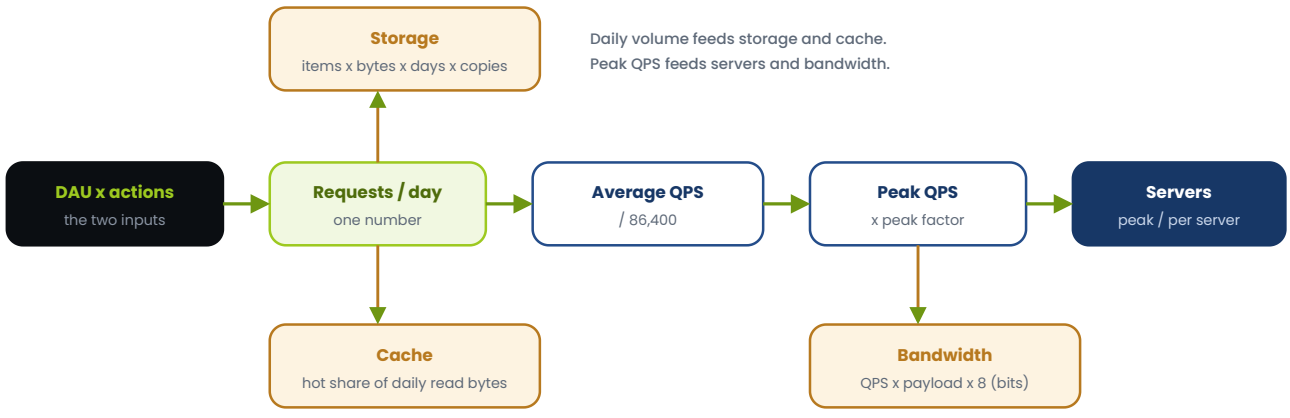
ONE WHITEBOARD, EVERY NUMBER

Sizing the short-link service

Say every assumption aloud first. The ones marked "new" are added in this lesson.

ASSUMPTION	VALUE
New links per month	100 million
Reads per write	100
Month length	30 days = 2,592,000 s
Peak factor (new)	2x average
Record size (new)	500 B (200 B without overhead)
Retention (new)	5 years
Replication (new)	3 copies
Redirect reply (new)	500 B
Hot data (new)	hottest 20% of daily read bytes
Cache hit rate (new)	90%
One app server (new)	2,000 simple requests/s

QUANTITY	CALCULATION	RESULT
Creates/s (avg)	100,000,000 / 2,592,000	38.6
Redirects/s (avg)	38.6 x 100	3,858
Creates/s (peak)	38.6 x 2	77
Redirects/s (peak)	3,858 x 2	7,716
Records, 5 years	100,000,000 x 60	6 billion
Storage (500 B)	6,000,000,000 x 500 B	3 TB
With 3 copies	3 TB x 3	9 TB
Short codes	62^7 vs 6 billion	0.17% used
Avg bandwidth	3,858 x 500 B x 8	15.4 Mbps
Reads per day	3,858 x 86,400	333 million
Daily read bytes	333,000,000 x 500 B	167 GB
Hot cache (20%)	0.20 x 167 GB	33 GB
DB reads at peak	7,716 x 0.10	772/s
App servers	7,716 / 2,000 = 4; x 2; + 1	9



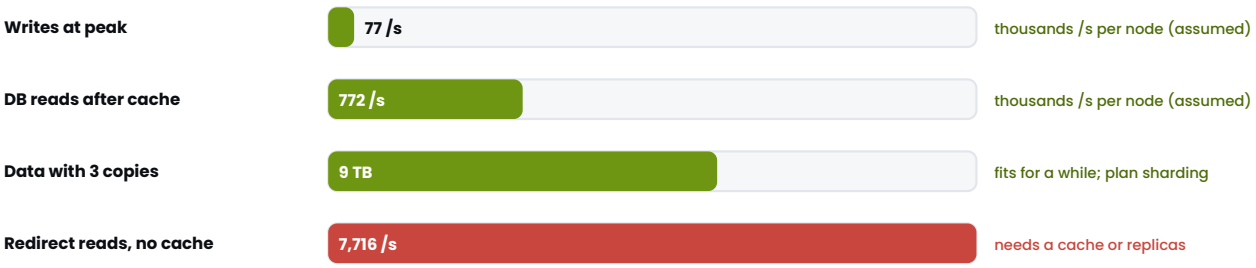
The estimation pipeline. Everything starts from the same two inputs, users and actions.

WHAT THE NUMBERS TELL YOU

From whiteboard to decisions

1. Reads dominate 100 to 1, so a **cache** in front of the database is justified, and the database drops from 7,716 to about 772 reads per second.
2. The hot set is about 33 GB, so the cache is **one or a few machines**.
3. 3 to 9 TB over five years is **plausible on one node for a while**: do not shard on day one, plan for it (Lesson 7).
4. About 31 Mbps at peak means redirects need **no CDN** for bandwidth reasons.
5. 77 writes per second at peak means **one primary** handles writes.
6. About **9 small app servers**, behind a load balancer (Lesson 5).

IS ONE MACHINE ENOUGH? COMPARE EACH ESTIMATE WITH ONE NODE (assumed capacities)



Is one machine enough? Only the uncached redirect load breaks it, and the cache fixes that.

THE 60-SECOND SPOKEN VERSION

"About 39 creates and 3,900 redirects per second on average, double at peak. Five years is 6 billion records, about 3 terabytes, 9 with three copies. Reads dominate 100 to 1, so I cache the hot 33 gigabytes and the database sees under 800 reads per second. Writes are under 80 per second. Bandwidth is about 15 megabits per second. Roughly 9 app servers with headroom."

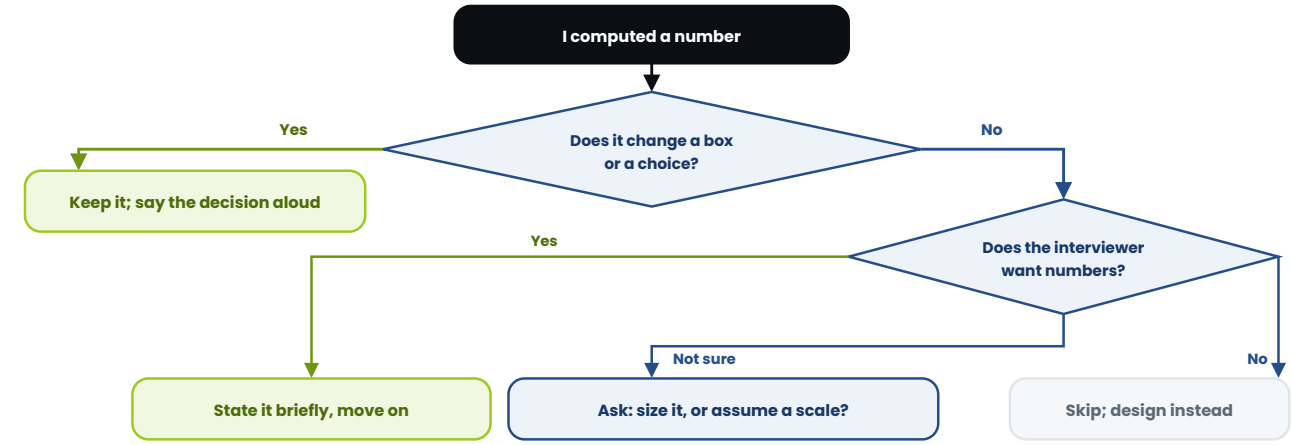
INTERVIEW MODE: SAY IT LIKE THIS

- "I size traffic, storage, bandwidth, cache and servers in that order and state each assumption. The numbers say: cache the hot set, one primary is enough, no CDN for redirects, and revisit sharding as storage grows."

ONE PICTURE TO REMEMBER

Is this estimate worth keeping?

Estimation earns its minutes only when it drives a decision or the interviewer asks for it.



Ask before you calculate when you are not sure the interviewer wants numbers.

STORY RECAP: ASHA, ONE MORE TIME

The interviewer asked "transactions per second?" Asha said: "UPI did 791 million a day in August 2026, that is about 9,000 per second, say 27,000 at a 3x peak. One database node will not take 27,000 writes a second safely, so I would queue and shard." **One number, one decision, thirty seconds.**

MEMORY HOOK: THREE NUMBERS TO TATTOO

86,400 seconds a day (about 10^5) · **1 million a day = 12/s** · **x 8** from bytes to bits.

THINGS THAT SOUND RIGHT AND ARE NOT

Common misconceptions

BELIEF	CORRECTION
"I need exact numbers."	Aim for the right order of magnitude. DesignGurus' example rounds 4,629 to about 5,000 and calls that a strong sentence [S4]; the primer says rounding aggressively is fine [S3].
"Average QPS is enough."	Systems fail at the peak. State a peak factor and why.
"Only senior interviews ask for this."	Mixed: Cisco asked QPS in a senior whiteboard [S6]; Blind posters disagree on Google, Facebook and others [S5]; a guide says juniors are more likely to be forgiven [X4]. Prepare it, but ask.
"Bits and bytes are the same."	A byte is 8 bits; links are in bits, storage in bytes. 200 MB/s is 1.6 Gbps.
"Storage is items times size."	Multiply by retention and replication too. 3 TB becomes 9 TB with three copies.
"The latency table is exact and current."	It is about 2012 and varies by version and hardware [S2, S1, S3]. Use ratios and orders of magnitude.
"Run servers as full as possible."	Keep headroom for spikes and failures; plan around half load and add a spare.
"Estimation is just maths for show."	A good estimate ends in a decision: cache, shard, queue, CDN.
"Stating assumptions looks unsure."	Stating them is the skill: an estimate is only as good as its stated assumptions.

WHAT GOES WRONG

Bonus: "I will do the arithmetic in my head silently, like a genius." The interviewer cannot grade a silent genius; they can only grade the visible steps.

INTERVIEW MODE: SAY IT LIKE THIS

- "I state assumptions, round to an order of magnitude, include peak, retention and replication, keep units straight, and finish with the design decision."

TEAR OUT THIS PAGE

One-page cheat sheet

WHAT THEY ASK	THE ANSWER	THE TRAP
Seconds in a day?	86,400 (about 10^5)	Using 3,600 or 24 x 60
1M requests a day in QPS?	About 12 per second	Forgetting to divide by 86,400
Average to peak QPS?	x a stated peak factor (2x, 3x, more)	Sizing for average
Storage formula?	Items x bytes x retention x replication	Leaving out retention or copies
MB/s to Mbps?	x 8	Mixing bits and bytes
Powers of two?	2^10 = KB, 2^20 = MB, 2^30 = GB, 2^40 = TB	Treating 1 GB as 10^6 bytes
Memory vs network vs disk?	100 ns, 500 µs (same DC), 10 ms seek	Quoting without units
How many servers?	Peak / per-server, headroom, spare	No per-server assumption
How big a cache?	Hot share of daily read bytes, e.g. 20%	Caching the whole dataset
Cross-continent RTT?	About 150 ms	Forgetting speed-of-light limits
Scale not given?	Ask; otherwise state an assumption and invite correction	Silently inventing numbers
Is the estimate useful?	Only if it changes a design choice	Computing for show

MEMORY HOOK: THE PIPELINE

DAU x actions then per day then / 86,400 then x peak then / per server.

MEMORY HOOK: STORAGE

I Bought Rice Cheap: Items x Bytes x Retention x Copies.

MEMORY HOOK: ORDER

The Smart Batch Cracks Systems: Traffic, Storage, Bandwidth, Cache, Servers.

WHAT YOU WILL ACTUALLY TYPE AT WORK

Where this shows up in real systems

Official statistics are anchors

NPCI publishes monthly UPI volumes; the Aug 2026 figures were reported on 1 Sep 2026 [S7]. Real numbers sanity-check assumptions.

Dashboards show QPS

Monitoring stacks (for example Grafana, named in a capacity guide [X3]) show requests per second, latency percentiles and errors, so measurements replace assumptions after launch.

wrk measures per-server capacity

From Lesson 2: wrk reports requests per second and latency percentiles, giving "2,000 per server" a real basis. Verify flags in wrk's docs.

ping shows round trips

ping -c 4 example.com prints RTT per probe. Compare with the 150 ms cross-continent figure.

Disk and memory checks

df -h (disk per mount), du -sh (folder size), free -h (memory, Linux) ground storage and cache estimates.

Cloud bills price egress

Providers charge for data leaving their network. Check the current price page rather than quoting from memory.

```
wrk -t4 -c100 -d30s http://localhost:8080/   # requests/s and latency percentiles
ping -c 4 example.com                        # illustrative output (invented): time=42.1 ms
df -h ; du -sh ./data ; free -h               # disk, folder size, memory
```

ASSUMPTION IN THIS LESSON	HOW A REAL TEAM REPLACES IT
2,000 requests/s per server	Load test one server with wrk; read requests/s at acceptable latency
Peak factor 2x or 3x	Peak vs average requests/s on the monitoring dashboard
500 B per record	Table size on disk / row count (df -h, du -sh)
90% cache hit rate	Hit and miss counters reported by the cache
40 ms user round trip	ping or browser timings from real users

INTERVIEW MODE: SAY IT LIKE THIS

- "In real teams I start with an estimate, then replace it with measured QPS and latency from dashboards and load tests."

FUNDAMENTALS WITH MODEL ANSWERS

Interview questions 1 and 2

Evidence is thin and mixed (see the honest note on page 2). Only the Cisco item is a dated first-person report; the rest are pattern-only or forum threads.

Q1 "What QPS does your system see, and how do you handle it?" SOURCE-CONFIRMED

Cisco, onsite whiteboard, Senior role, Jul 2026. Interviewers asked about QPS, rate limiting, circuit breaker, queues, sync vs async, REST vs gRPC vs GraphQL [S6].

Model answer. "About 3,500 requests per second on average: 30 million daily users doing 10 actions each is 300 million a day, divided by 86,400. I assume a 3x peak, so about 10,500. One server handles about 2,000 requests per second, so I need 6 at peak and about 12 with headroom, plus a spare. Reads dominate, so a cache absorbs most of them."

Why does QPS matter? It decides capacity: how many servers, whether one database node is enough, where queues or caches go.

Peak is double what you assumed? About 21,000. At 2,000 per server about 11, so about 22 with headroom; re-check cache and database.

DB tops out at 5,000 reads/s? With a 90% hit rate the DB sees about 2,100 at the 21,000 peak, which fits. Without the cache it would not.

Weak: "A lot, we would scale horizontally." ("A lot" is not a unit.)

Strong: "About 3,500 average, 10,500 at peak, and here is the server and cache count."

Q2 Estimate the traffic and storage for a URL shortener PATTERN-ONLY

No verbatim estimation question found. Context: Amazon asked to design a URL shortener in a four-round loop, Jun 2025 [S8]; a Seattle candidate, May 2025, reported "a scalable URL shortening service like Bitly" [X8].

Model answer. "I assume 100 million new links a month and 100 reads per write. That is about 39 creates and 3,900 redirects per second on average, double at peak. Five years is 6 billion records at about 500 bytes, 3 terabytes, 9 with three copies. A 7-character base-62 code has 3.5 trillion values, enough."

Why 100:1? Every click is a redirect, creation happens once. I would confirm the ratio.

200 B per record? 6 billion x 200 B = 1.2 TB, 3.6 TB with three copies.

What does the number change? Peak reads of 7,716/s justify a 33 GB cache; 77 writes/s means one primary is enough.

Weak: "It depends on users."

Strong: "Assumptions, the chain of arithmetic, and the decisions."

FUNDAMENTALS WITH MODEL ANSWERS

Interview questions 3 to 5

Q3 Is estimation required? What if no scale is given? CANDIDATE-REPORTED

Unnamed companies, anonymous Blind thread, undated; posters disagree [S5]. An Alex Xu-style summary advises asking whether estimation is required [X5].

Model answer. "I ask whether they want me to size the system. If no scale is given, I state assumptions, such as 30 million daily users, and invite correction. I keep it to a few minutes and make sure each number leads to a decision."

> **Why would some not ask?** Some grade architecture and reasoning mainly; at junior levels weak estimation may be forgiven if the architecture is solid [X4].

> **"Assume a billion users."** Assume 20% daily active: 200 million DAU, then actions per user, per day, QPS. A billion is registered users, not load.

> **They disagree with your number.** Ask what they expect, adjust, recompute; the method matters more than the first value.

Weak: "I skip numbers, they are boring."

Strong: "I ask, assume, compute briefly, and tie it to a decision."

Q4 Convert DAU into average and peak QPS PATTERN-ONLY

No company found. A prep guide gives the recipe "DAU x actions = daily requests; divide by 100K for average QPS; multiply by a peak factor" [X1].

Model answer. "Requests per day equals DAU times actions per user. Divide by 86,400, or about 100,000 for quick mental maths, for average QPS. Multiply by a stated peak factor, such as 2 to 3, for peak."

> **Why not size for average?** Systems fail at peaks; a sale or evening rush can be several times the average.

> **200M DAU, 2 writes and 200 reads each?** 400 million writes a day is about 4,630/s; 40 billion reads about 463,000/s; 100:1 [S4].

> **What does 100:1 imply?** Optimise the read path: caches, read replicas, maybe a CDN for static content.

Weak: "Users times something."

Strong: "DAU x actions / 86,400 x peak factor, each assumption stated."

Q5 Estimate storage for five years, including replication PATTERN-ONLY

No company found. The five-step capacity worksheet pattern appears in guides [X2].

Model answer. "Items per day times bytes per item gives daily storage. Multiply by retention days and the replication factor. 100 million records a month at 500 bytes is 50 GB a month, 3 TB in five years, 9 TB with three copies."

> **Why multiply by replication?** Each copy is stored separately: durability and read capacity at three times the disk.

> **200 B, 10 million a day, 1 year, 3 copies?** 2 GB a day, 730 GB a year, 2.19 TB with three copies.

> **Too big for one node?** 2 to 9 TB can fit on one machine depending on hardware (assumption), but with replicas and growth plan for sharding (Lesson 7).

Weak: "Items times size."

Strong: "Items x size x retention x copies, and the design consequence."

FUNDAMENTALS WITH MODEL ANSWERS

Interview questions 6 to 8

Q6 Compare memory, SSD, network and disk latency PATTERN-ONLY

No company found. Reference: the latency table credited to Norvig and Dean [S2, S1].

Model answer. "Main memory is about 100 ns, a same-data-centre round trip about 500 µs, an SSD random read about 150 µs, a disk seek about 10 ms, a cross-continent round trip about 150 ms. Each boundary costs one to three orders of magnitude."

> **Why does it matter?** Keep hot data in memory and cut network round trips.
>
> **Thirty 256 KB thumbnails, serial vs parallel?** Serial about 560 ms, parallel about 18 ms [S1].
>
> **These are from 2012. Still right?** Order of magnitude only. Versions differ and devices vary; measure your own system [S2].

Weak: A list of numbers without units.

Strong: The ratios, one example and the caveat.

Q7 How many servers do you need? PATTERN-ONLY

No verbatim question; the Blind thread shows some interviewers asked for the number of servers [S5] (candidate-reported).

Model answer. "Peak QPS divided by what one server handles, then headroom and a spare. With 7,716 peak requests per second and 2,000 per server, that is 4 servers, 8 at 50% load, 9 with a spare. I would confirm per-server capacity with a load test."

> **Why headroom?** Servers slow sharply near full load; spikes and failures need slack.
>
> **Requests take 20 ms: in flight at peak?** Little's Law: 7,716 x 0.02 s = about 154.
>
> **A server dies.** With a spare and a load balancer the rest absorb it; Lesson 5 covers health checks.

Weak: "A few servers."

Strong: "The formula, the per-server assumption, the headroom, and how I would verify it."

Q8 Estimate bandwidth for an image or video service PATTERN-ONLY

No company found.

Model answer. "Bandwidth is QPS times payload, converted to bits. 1,000 image views per second at 200 KB is 200 MB/s, which is 1.6 Gbps. One 1 Gbps link carries about 100 to 125 MB/s, so I need more links or, better, a CDN to serve most images from the edge."

> **Why does it matter if storage fits?** The network can be the bottleneck and egress is billed.
>
> **5 million uploads a day at 300 KB?** 1.5 TB a day, about 2.7 PB over five years before copies.
>
> **How do you cut the egress?** A CDN, smaller image variants and compression; each trades cost for complexity.

Weak: "Add more bandwidth."

Strong: "Compute, convert to bits, compare with a link, then choose a CDN."

HOW TO THINK WHEN IT BREAKS

Scenario questions 1 to 4

Labels refer to the underlying situation; the steps and all numbers are original assumptions.

1. "Design it for 100 million users." Nothing else.

PATTERN-ONLY

1. Ask: registered or daily active? What do they do, and how often?
2. If no answer: "Say 20% active daily, so 20 million DAU, 10 actions each."
3. Compute: 200 million a day; / 86,400 = about 2,300/s; x 3 = about 7,000 at peak.
4. Decision: "A few servers and a cache; no exotic infrastructure yet."

2. Profile photos for 20 million users?

PATTERN-ONLY

1. Assume one 2 MB original and one 100 KB thumbnail each (assumptions).
2. 20,000,000 x 2.1 MB = 42,000,000 MB = 42 TB.
3. With 3 copies: 126 TB.
4. Lever: smaller originals or compression halves it.

3. You say 5 PB; the interviewer says too high.

PATTERN-ONLY

1. Do not defend; recompute aloud. Ask which assumption looks wrong.
2. Re-check items, bytes, retention, replication and units (MB vs GB).
3. Compare: UPI's whole volume is about 791 million transactions a day [S7]; if you claim more, an assumption is inflated.
4. Fix and recompute: 5 MB per item instead of 500 KB was 10x; the answer drops to 500 TB.

4. Fine at average, fell over at the 12:00 sale.

PATTERN-ONLY

1. Average 1,000/s; sale peak 10x = 10,000/s.
2. Sized for 1,000 at 2,000 per server: 1 server. At 10,000/s you needed 5, or 10 with headroom.
3. Fixes: size for peak, autoscaling or queues, cache hot product pages.
4. Next time ask for peak-factor evidence (past sale traffic).

HOW TO THINK WHEN IT BREAKS

Scenario questions 5 to 8

5. High bill: images served from origin at about 3 Gbps. PATTERN-ONLY

1. 3 Gbps / 8 = 375 MB/s.
2. Per day: 375 MB x 86,400 = about 32 TB leaving the system.
3. Images are static: a CDN (Lesson 5) serves most from the edge; image variants shrink sizes.
4. Check current egress prices; none given here because they change.

6. Cache for a 1 TB dataset where a small part is hot. PATTERN-ONLY

1. Ask the hot fraction. Suppose 5% gets most reads (assumption).
2. Hot set: 0.05 x 1 TB = 50 GB.
3. 50 GB with headroom fits a few machines; 950 GB stays in the DB.
4. Measure the real hit rate after launch and adjust.

7. A page must respond within 200 ms. Feasible? PATTERN-ONLY

1. User RTT 40 ms (assumed) + TLS one more RTT: 80 ms before the request arrives (a reused connection removes this).
2. Backend: 5 calls in the data centre at 0.5 ms = 2.5 ms, one SSD read 0.15 ms [S2], 20 ms app work = about 23 ms.
3. Total: 80 + 40 (request RTT) + 23 = about 143 ms; 57 ms spare.
4. The network dominates; shaving the SSD read changes nothing.

8. Food-delivery orders per second at lunch in one city. PATTERN-ONLY

1. 10 million people, 2% order a day: 200,000 orders (invented assumptions).
2. 2-hour lunch peak carries 25%: 50,000 in 7,200 s = about 7 orders/s.
3. About 30 API calls per order (browse, cart, pay, track): about 210 calls/s.
4. Orders are small; browsing and tracking traffic dominate. Only seven biryanis a second.

THE PATTERN BEHIND EVERY SCENARIO

Ask the missing number, **assume** it aloud if nobody answers, **compute** in small steps with units, and **say the decision** the number supports.

SOLVE THESE BEFORE YOU LOOK

Practice problems 1 to 4

Original problems, not reported questions. Cover the solution, try it on paper, then check.

P1 Requests to QPS EASY

Average QPS for (a) 10 million requests a day; (b) 1 billion a day; (c) 50 million a month (30 days); (d) 5,000 per second into requests per day.

Hint. Day = 86,400 s. Month (30 days) = 2,592,000 s.

SOLUTION

(a) 10,000,000 / 86,400 = about 116/s. (b) 1,000,000,000 / 86,400 = about 11,574/s. (c) 50,000,000 / 2,592,000 = about 19.3/s. (d) 5,000 x 86,400 = 432,000,000 a day.

P2 Sizes and units EASY

(a) 1 billion records of 100 B? (b) 50 million users with a 2 KB profile? (c) 8 Mbps in MB/s? (d) 1.6 Gbps in MB/s?

Hint. Use thousand, million, billion. A byte is 8 bits.

SOLUTION

(a) 10^9 x 100 B = 10^11 B = 100 GB. (b) 50,000,000 x 2 KB = 100,000,000 KB = 100 GB. (c) 8 / 8 = 1 MB/s. (d) 1,600 Mbps / 8 = 200 MB/s.

P3 A chat app NUMERICAL

10 million DAU, 40 messages each a day, 100 B per message, kept 1 year, 3 copies, peak 3x. Find average and peak messages/s, storage per day, and for the year with copies.

Hint. Messages per day = DAU x 40.

SOLUTION

400,000,000 messages a day; average about 4,630/s; peak about 13,900/s. Storage: 40 GB a day; x 365 = 14.6 TB; x 3 = 43.8 TB.

P4 An image-sharing service NUMERICAL

5 million uploads a day at 300 KB, read:write 20:1, 5 years, no copies. (a) Storage per day and for 5 years. (b) Views per day and daily outbound data. (c) Average bandwidth in MB/s and Gbps. (d) 1 Gbps links (125 MB/s) at average and at a 3x peak?

Hint. Views = uploads x 20. Daily data / 86,400 = bytes per second; x 8 for bits.

SOLUTION

(a) 1.5 TB a day; x 365 x 5 = 2,737.5 TB = about 2.7 PB. (b) 100 million views; 30 TB a day out. (c) 30 TB / 86,400 = about 347 MB/s = about 2.78 Gbps. (d) Average 347 / 125 = 2.8, so 3 links; peak about 1,040 MB/s = 8.3, so 9 links. This is the number that argues for a CDN.

SOLVE THESE BEFORE YOU LOOK

Practice problems 5 and 6

P5 Thumbnails again, with the table MEDIUM

Using the 2012 table [S2] (SSD random read 150 µs, 1 MB from SSD 1 ms, disk seek 10 ms, 1 MB from disk 20 ms), estimate 30 images of 256 KB loaded serially from (a) SSD and (b) disk. What changes in parallel?

Hint. Time per image = access + 256 KB / transfer speed. 256 KB is a quarter MB.

SOLUTION

(a) SSD: 0.15 + 0.25 = about 0.4 ms each; serial 12 ms. (b) Disk: 10 + 5 = 15 ms each; serial 450 ms, about 37x slower than SSD. In parallel the total approaches one image: about 0.4 ms or 15 ms, before contention. It differs from Dean's 560 ms because table versions use different transfer rates (30 vs 50 MB/s) [S1, S2]; the lesson is the ratio and the benefit of parallelism.

P6 Notifications at UPI scale HARD

791 million transactions a day [S7]; 2 notifications each, peak 3x, 300 B each, one worker sends 500/s. (a) Average and peak per second? (b) Workers at peak and with 2x headroom? (c) Peak bandwidth in Mbps? (d) If the true peak is 5x? (e) One sanity check.

Hint. Notifications per day = 791,000,000 x 2. Workers = peak / 500.

SOLUTION

(a) 1,582,000,000 a day; about 18,310/s average; about 54,930/s peak. (b) About 110 workers; 220 with headroom. (c) 54,930 x 300 B = 16.5 MB/s = about 132 Mbps. (d) 91,550/s, about 183 workers, 366 with headroom, about 220 Mbps: workers scale linearly with the peak factor, so justify it. (e) 18,310/s is twice the 9,155 transactions/s, as assumed; 132 Mbps is small next to one 1 Gbps link.

HOW TO CHECK YOUR OWN ANSWER

Write the assumption table first. Then one line per step, each with its unit. Then a sanity check against something real. Then the decision. If any of the four is missing, the answer is incomplete, even if the number is right.

TEN QUESTIONS, TWO MINUTES EACH

Mini quiz

Original questions. Cover the answers, answer all ten, then check.

1. How many seconds are in a day?
A) 3,600 B) 86,400 C) 1,440 D) 100,000 exactly
Answer B. 24 x 60 x 60. A is an hour; C is minutes in a day; D is a rounding shortcut.

2. About how many requests/s is 10 million a day?
A) 12 B) 116 C) 1,160 D) 11,600
Answer B. 10,000,000 / 86,400 = about 116. A is for 1 million a day.

3. 2^30 bytes is approximately:
A) 1 million B) 1 billion C) 1 trillion D) 1 thousand
Answer B [S3]. 1,073,741,824: a gigabyte.

4. 6 billion records of 500 B, 3 copies?
A) 3 TB B) 9 TB C) 1.2 TB D) 90 TB
Answer B. 3 TB x 3. A forgets replication; C uses 200 B and no copies.

5. 2 MB per second is how many Mbps?
A) 0.25 B) 2 C) 16 D) 200
Answer C. x 8.

6. Same-DC round trip (500 µs) vs memory (100 ns)?
A) 5x B) 50x C) 500x D) 5,000x
Answer D [S2]. 500,000 ns / 100 ns.

7. Dean's thumbnails: which wins and by how much?
A) Serial, 10x B) Parallel, about 560 ms vs 18 ms C) Equal D) Serial
Answer B [S1]. Thirty serial 10 ms seeks dominate.

8. Average 1,000 QPS, peak factor 3. Size for:
A) 1,000 B) 3,000 C) 300 D) 333
Answer B. Systems fail at peak.

9. No scale given. Best first move?
A) Refuse B) Ask, then state assumptions and invite correction C) Random huge number D) Skip numbers
Answer B [S4, X5].

10. 8,000 req/s, 50 ms each. In flight (Little's Law)?
A) 160 B) 400 C) 4,000 D) 8,000
Answer B. 8,000 x 0.05 = 400.

FIVE SENTENCES TO KEEP

Recap and the 60-second answer

1. An estimate is a stated-assumption calculation to the right order of magnitude; its job is to decide the design.
2. Users and actions give requests per day; / 86,400 gives average QPS; a stated peak factor gives peak; servers = peak / per-server capacity, plus headroom and a spare.
3. Storage = items x bytes x retention x replication; bandwidth = QPS x payload x 8 for bits; cache = hot share of daily read bytes.
4. Latency numbers are about 2012 and approximate: memory 100 ns, DC round trip 500 µs, disk seek 10 ms, cross-continent 150 ms. Use ratios.
5. Interviewers differ, so ask, state assumptions, sanity-check against something real, and end each number with a design decision.

SAY THIS IN AN INTERVIEW: "HOW WOULD YOU ESTIMATE THE LOAD FOR THIS SYSTEM?"

"I would ask whether you want me to size it, and if no scale is given I would state assumptions. Daily active users times actions per user gives requests per day; I divide by 86,400 for average QPS and multiply by a peak factor I would justify, say 2 to 3. Storage is items per day times bytes per item times retention times the replication factor. Bandwidth is QPS times payload, times eight for bits. For the cache I take the hot share of daily read bytes. Servers are peak QPS divided by what one server handles, with headroom and a spare, which I would confirm with a load test. I round aggressively, keep units straight, sanity-check against something real, and then say what the numbers change: for example, 100 to 1 reads means a cache, and a few thousand writes per second means a single primary may not be enough."

CHECK OUR WORK

Sources

S = opened in full; X = seen only in search extracts. Every calculation in this lesson was recomputed line by line.

ID	PAGE	URL
S1	High Scalability: Google Pro Tip, back-of-the-envelope calculations (26 Jan 2011)	highscalability.com/google-pro-tip-use-back-of-the-envelope-calculations-to-choo/
S2	GitHub gist: Latency Numbers Every Programmer Should Know (Jonas Bonér; credits Dean and Norvig)	gist.github.com/jboner/2841832
S3	System Design Primer (Mintlify mirror): Back-of-the-Envelope Calculations	mintlify.com/donnemartin/system-design-primer/interview/calculations
S4	DesignGurus: Four-Step Framework (updated May 2026)	designgurus.io/system-design-interview/framework
S5	Blind: Is back of the envelope estimation expected/required? (undated)	teamblind.com/post/is-back-of-the-envelope-estimation-expectedrequired-in-a-system-design-interview-8hhizas2
S6	Glassdoor: Cisco Senior SWE reports (incl. Jul 2026)	glassdoor.com.br/Entrevista/Cisco-Entrevista-E1425-RVW103576482.htm
S7	Free Press Journal: UPI Transactions Hit Record 24.51 Billion In August (1 Sep 2026)	freepressjournal.in/business/upi-transactions-hit-record-2451-billion-in-august-as-digital-payments-continue-rapid-growth
S8	Glassdoor: Amazon Embedded SDE (interview Jun 2025)	glassdoor.co.in/Interview/Amazon-Interview-E6036-RVW102556955.htm
X1	GitGood: top system design questions 2026 (DAU x actions, / 100K)	gitgood.dev/blog/top-30-system-design-interview-questions-2026
X2	DesignGurus Substack: 5-step capacity estimation worksheet	designgurus.substack.com/p/the-5-step-capacity-estimation-worksheet
X3	GeeksforGeeks: Capacity estimation in systems design	geeksforgeeks.org/system-design/capacity-estimation-in-systems-design/
X4	System Design Handbook: back-of-the-envelope calculation	systemdesignhandbook.com/guides/back-of-the-envelope-calculation/
X5	Scribd: System Design Inshort (Alex Xu-style summary)	scribd.com/document/881369248/System-Design-Inshort
X7	Latency-table variant from the Google SRE book	wiki.liujiacai.net/cheatsheets/latency-numbers/
X8	Glassdoor (Italy mirror): Amazon SDE-I, May 2025, URL shortener prompt	static.glassdoor.it/Interview/Amazon-Software-Development-Engineer-SDE-I-Interview-Questions-EI_IE6036.0,6_KO7,42_IP50.htm

SELF-CHECK

Can you say yes to all four?

I can turn users into QPS
Average and peak, out loud, in under a minute.

I can size storage
With retention and replication, in the right unit.

I can size bandwidth and cache
Bits vs bytes, and the hot set only.

I can end with a decision
Cache, shard, queue, CDN, or "one machine is enough".

Covered here, and where else

Units, QPS, storage, bandwidth, cache, latency and decisions: **fully**. Server counts and Little's Law: introduced (Lesson 5). Turning estimates into designs: Lessons 24 to 31.

Not covered (interviewers sometimes go here)

- Cost estimation in rupees or dollars (prices change; check the provider)
- Latency percentiles and queueing theory beyond Little's Law
- Capacity for ML and LLM workloads (tokens/s, GPU counts)
- Fan-out multipliers and write amplification (Lessons 3 and 27 touch them)
- Burst modelling, multi-region splits, compression ratios
- Real hardware specs: per-server capacity, peak factors and record sizes here are assumptions

> "Roughly how many per second?" **Now you answer with a number, a unit, an assumption and a decision.**

YOUR NEXT STEPS

STEP 01
Solve problems 1 to 6 on paper

STEP 02
Submit SD4-1 on the daily form

STEP 03
Read ahead: Lesson 5, scaling